

Digital Assistant (Agentic AI)

Production-style serverless AI assistant with tools, memory, async workers, and quota governance.

Live App: <https://digital-assistant.agentaigr.site>

Deployment: Terraform + GitHub Actions

Architecture: AWS Serverless

Problem

- Most chat demos break in production: timeout failures, unreliable tool outputs, and missing usage controls.
- Long-running tasks exceed synchronous API limits and degrade UX.
- Without verifiable actions and observability, operations and trust become fragile.

Solution & Core Features

- Tool-enabled agent: web search, PDF export, email, file upload parsing, memory extraction.
- Async execution: /chat returns job_id; worker completes long tool chains.
- Daily quota enforcement: tokens/PDF/email tracked server-side with deterministic limits.
- User-approved memory: candidate memories require explicit approval before reuse.

- Reliability guardrails: tool-claim validation and safe fallbacks for failed tool actions.
- Upload path optimized for production: browser direct-to-S3 via presigned URLs.
- Responsive UI tuning for mobile/tablet chat header actions and quota visibility.
- Observability via OpenTelemetry instrumentation (traces/logs) exported to Sentry via OTLP.
- Split domains in prod: website on CloudFront domain, API on API Gateway custom domain.

Architecture (High Level)

- Frontend: Next.js served from S3 + CloudFront.
- API: FastAPI running in AWS Lambda behind API Gateway REST.
- Worker: dedicated Lambda for tool-heavy asynchronous runs.
- State: S3 for artifacts/memory, Upstash Redis for jobs and quotas.
- Observability: OpenTelemetry for API + worker, exported to Sentry (OTLP); CloudWatch for AWS runtime logs/metrics.
- Infra as code: Terraform manages IAM, ECR, Lambda, API Gateway, S3, CloudFront, Route53, ACM.

Stack

Backend
Python, FastAPI, Agent runtime, MCP

Frontend
Next.js, React, TypeScript

Cloud
AWS Lambda, API Gateway, S3, CloudFront, ECR, IAM, Route53, ACM

Data & Ops
Upstash Redis, OpenTelemetry/Sentry, CloudWatch, GitHub Actions